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Studierichting: Informatica
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$$(-1 + i)^{10}$$

$$\arg(z): \tan \alpha = \frac{1}{-1} = -1 \Rightarrow \alpha = \frac{\pi 3}{4}$$

$$\text{modulus: } r = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$$

$$\text{goniometrische vorm: } \sqrt{2} \operatorname{cis} \frac{\pi 3}{4}$$

$$\left(\sqrt{2} \operatorname{cis} \frac{\pi 3}{4} \right)^{10} = (\sqrt{2})^{10} \operatorname{cis} \frac{10\pi 3}{4}$$

$$= 32 \operatorname{cis} \frac{15\pi}{2} = 32 \operatorname{cis} \frac{3\pi}{2}$$

$$= -32i$$

References